DATA SCIENCE & ARTIFICIAL INTELLIGENCE



Probability

Lecture No. 06



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Recap of previous lecture



Topic

Probability (Part 05)

(BAYE'S Theosem)

Topics to be Covered



Topic

BASICS of PROBABILITY (Part-06)

1. Remaining Questions of Bayes's Th.
2. Concept of with & w/o Replacement
3. Questions Based on Playing Cards

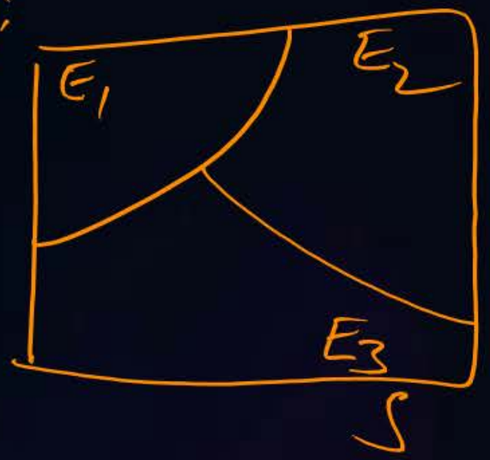
M.E Events :- E_1, E_2, E_3 are called ME if $E_i \cap E_j = \phi \ \forall i \neq j$

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3)$$

$$\boxed{P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)} \quad -0-0-0+0$$

Exhaustive Events :- E_1, E_2, E_3 are called Exhaustive if $\boxed{E_1 \cup E_2 \cup E_3 = S}$

Note :



By Diagram we can conclude that E_1, E_2, E_3 are ME & Exhaustive

Sp Note :- if E_1, E_2, E_3 are ME & Exhaustive Events, then

$$\boxed{P(E_1) + P(E_2) + P(E_3) = 1}$$ learn.

Proof: $P(E_1 \cup E_2 \cup E_3) = P(S) \Rightarrow P(E_1) + P(E_2) + P(E_3) = 1$ Hence Proved

Eg: $S = \{1,2,3,4,5,6\}$ & let $E_1 = \{1,2\}$, $E_2 = \{3\}$, $E_3 = \{4,5,6\}$

$\Rightarrow P(E_1) = \frac{2}{6}$, $P(E_2) = \frac{1}{6}$, $P(E_3) = \frac{3}{6}$

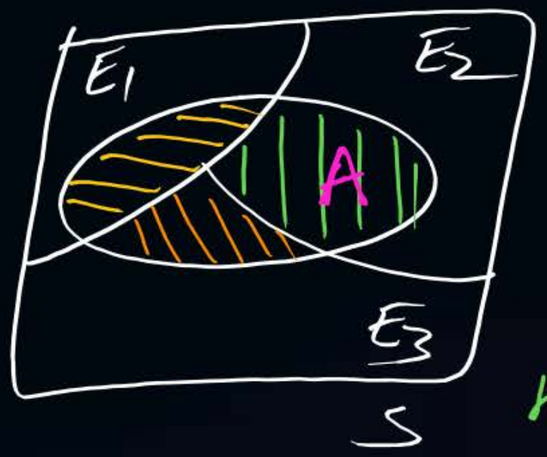
∵ $E_1 \cup E_2 \cup E_3 = S$ & E_1, E_2, E_3 are Exhaustive

∵ $E_1 \cap E_2 = E_2 \cap E_3 = E_3 \cap E_1 = \phi$ i.e. $E_i \cap E_j = \phi \; \forall i \neq j$ & these are ME also.

Hence $P(E_1) + P(E_2) + P(E_3) = \frac{2}{6} + \frac{1}{6} + \frac{3}{6} = \frac{6}{6} = 1$ Hence verified

Law of Total Probability

Let E_1, E_2, E_3 are ME and Exhaustive events, and A is an Event which can occur with all then



$$P(A) = P(E_1)P(A/E_1) + P(E_2)P(A/E_2) + P(E_3)P(A/E_3) \quad \text{---(1)}$$

Here E_1 & A are not ME
E_2 & A " " "
E_3 & A " " "

But E_1, E_2, E_3 are M.E

Baye's Th.

Theory same as above. then

$$P(E_1/A) = \frac{P(E_1)P(A/E_1)}{P(A)}, \quad P(E_2/A) = \frac{P(E_2)P(A/E_2)}{P(A)}, \quad P(E_3/A) = \frac{P(E_2)P(A/E_3)}{P(A)}$$

(Inverse Prob Th)

Note ① $P(A/E_1)$ = Prob of A, when E_1 has already occurred

$P(E_1/A)$ = Prob of E_1, when A has " "

② N. Cond^n for the existence of Baye's Th & Law of Total Prob is "Associated Events must be ME & Exhaustive"

③ In Baye's Th; $A = \{$ Assume that event as A which is given as Condition $\}$

& In Law of Total Prob, $A = \{$ Assume that Event at A which is Required $\}$

④ To solve tough Questions of Conditional Prob, we will use Baye's Th.

⑤ Whenever Question is Based on Conditional Prob & there is a feeling of Cross Check something then we should use B-Th.

⑥ If we have no Condition in Quest then we should use Law of Total Prob.

HW8 A person is known to speak truth 3 out of 4 times. He throw a die and Reports that it is six then Find the prob that it is actually six?

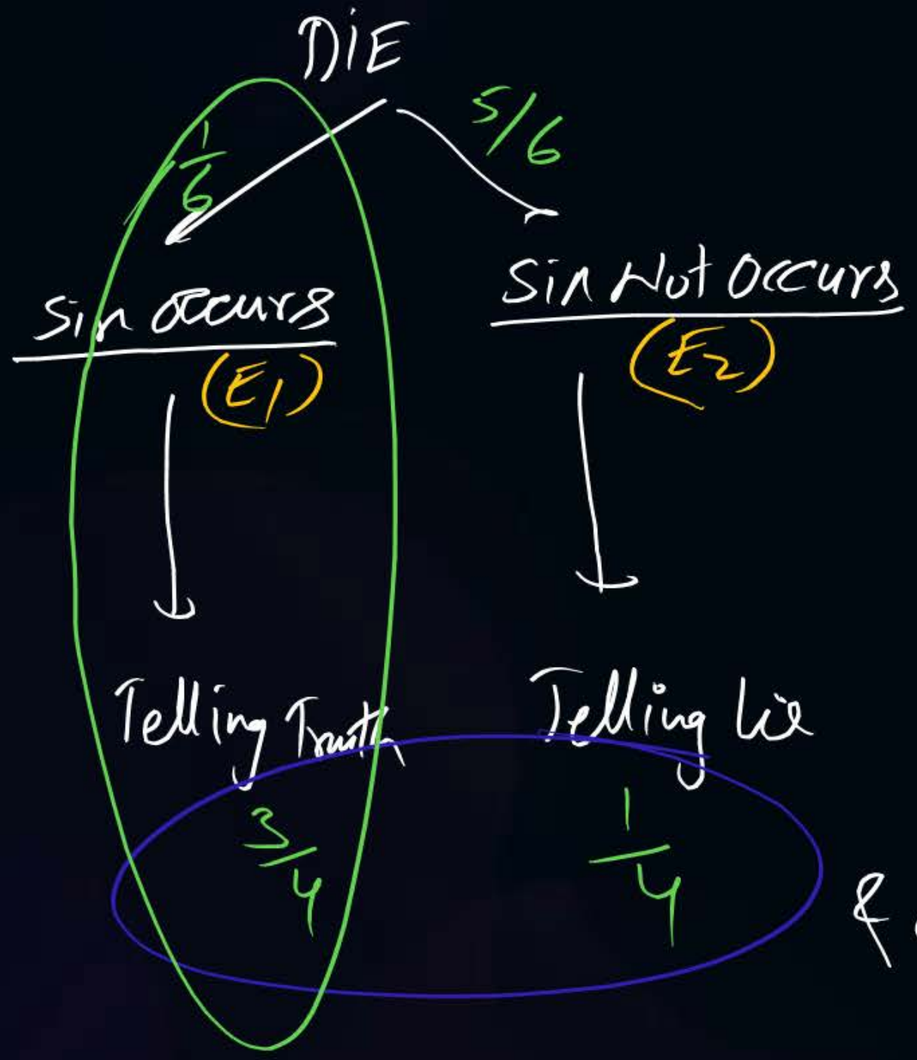
Sol: $P(\text{Telling Truth}) = \frac{3}{4}$, $P(\text{Telling Lie}) = \frac{1}{4}$

$A = \{\text{Man Reports that it is six}\}$

$E_1 = \{\text{six occurs}\} = \{6\} \Rightarrow P(E_1) = \frac{1}{6}$

$E_2 = \{\text{six Not occurs}\} = \{1,2,3,4,5\} \Rightarrow P(E_2) = \frac{5}{6}$

$A/E_1 = \{\text{Man is telling truth}\} = \frac{3}{4}$, $A/E_2 = \{\text{Man is telling Lie}\} = \frac{1}{4}$



Man Reports that it is Sin

A :

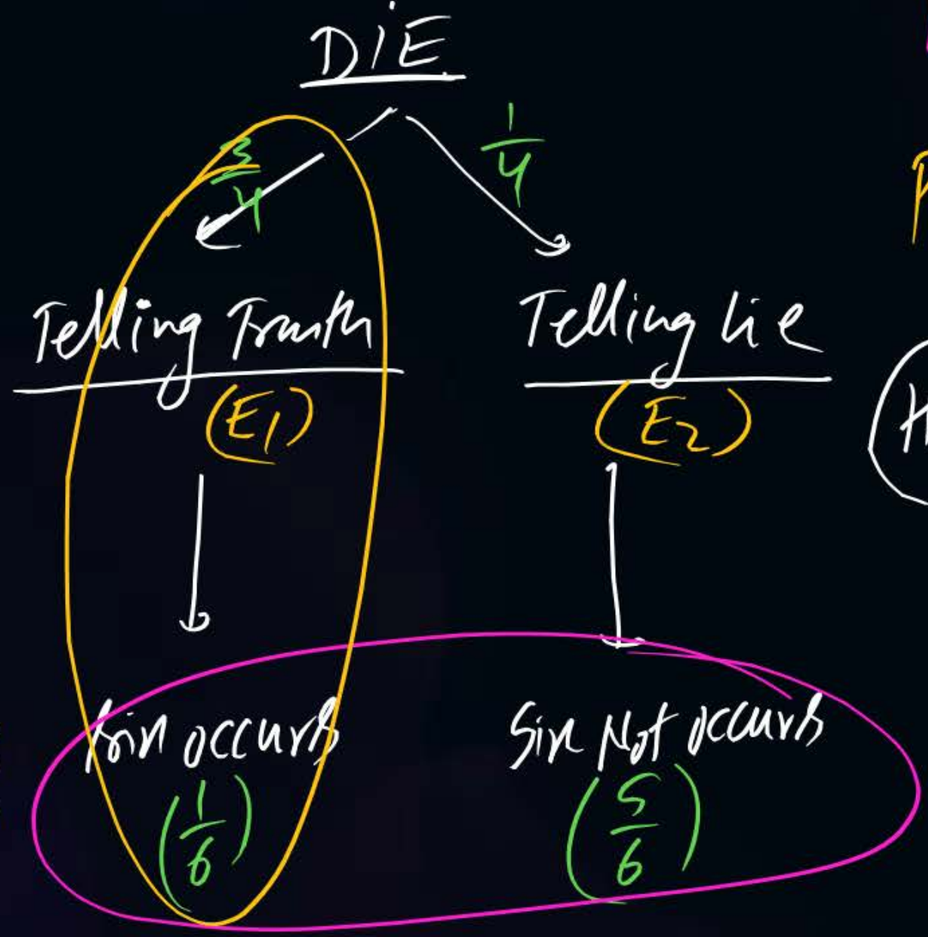
$$P(A) = \left(\frac{1}{6} \times \frac{3}{4}\right) + \left(\frac{5}{6} \times \frac{1}{4}\right) = \frac{8}{24}$$

$$P(\text{actually Sin}) = P(E_1/A) = \frac{\frac{1}{6} \times \frac{3}{4}}{8/24} = \frac{3}{8}$$

i.e out of 24 Reports given by them 8 reports representing that "It is Sin" & out of 8 Reports 'Represeting Sin' only 3 are Correct.

M-II

$A = \{$ Man Reports that it is Six $\}$



$$P(A) = \frac{3}{4}\times\frac{1}{6} + \frac{1}{4}\times\frac{5}{6} = \frac{8}{24}$$

$$P(E_1/A) = \frac{\frac{3}{4}\times\frac{1}{6}}{8/24} = \frac{3}{8}$$

(Here, E_1 & E_2 are ME as well as Exhaustive)

Q: In a town 10% of the population is COVID +ve. A new diagnostic kit arrives in the market. This kit correctly identifies COVID +ve individual 95% of time & COVID -ve individual 89% of time.

A person is tested by this kit and is found to be +ve then find the prob that person is actually +ve? (Ans = 0.48)

Condition

Sol: $A = \{$ Person is declared +ve by kit $\}$

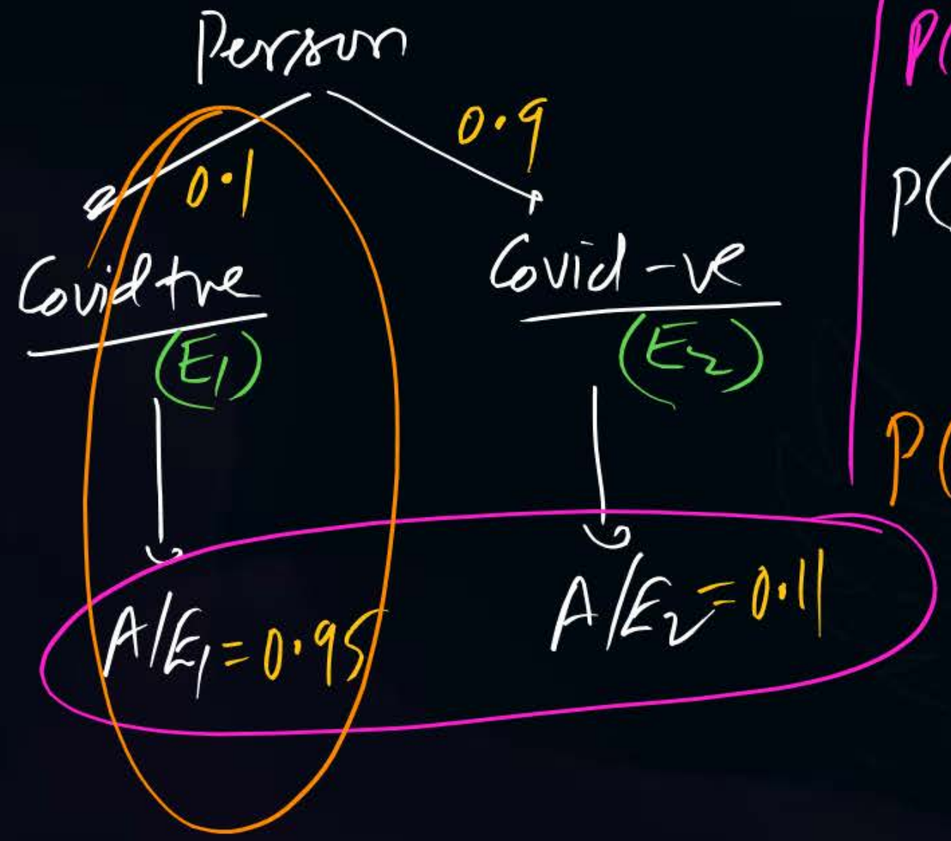
$E_1 = \{$ Covid +ve $\} \Rightarrow P(E_1) = 0.1$

$E_2 = \{$ Covid -ve $\} \Rightarrow P(E_2) = 0.9$

+ve person → 95% True, 5% False & -ve person → 89% True, 11% False

$A/E_1 = \left\{\frac{\text{Declared +ve by kit}}{\text{Covid +ve}}\right\} = \{\text{kit is giving True Result for +ve Person}\} \Rightarrow 0.95$

$A/E_2 = \left\{\frac{\text{Declared +ve by kit}}{\text{Covid -ve}}\right\} = \{\text{kit is giving false Result for -ve person}\} \Rightarrow 0.11$



Person Declared +ve by kit A :

$P(A) = 0.1 \times 0.95 + 0.9 \times 0.11 = 0.194$

$P(\text{person is declared +ve by kit}) = \frac{194}{1000}$

$P(\text{actually +ve}) = P(E_1/A)$

$= \frac{0.1 \times 0.95}{0.194} = \frac{95}{194}$

Analysis ① $P(A) = \frac{194}{1000}$, $P(E_1/A) = \frac{95}{194}$

i.e out of 1000 persons tested by kit, 194 are declared +ve by kit

& out of 194 +ve Declared person by kit, 95 are Actually +ve

② 1000 persons → 194 Declared +ve by kit → (95) are actually +ve (True Result)
→ 99 are actually -ve
→ 806 Declared -ve by kit → 801 are actually -ve (True Result)
→ (5) are actually +ve

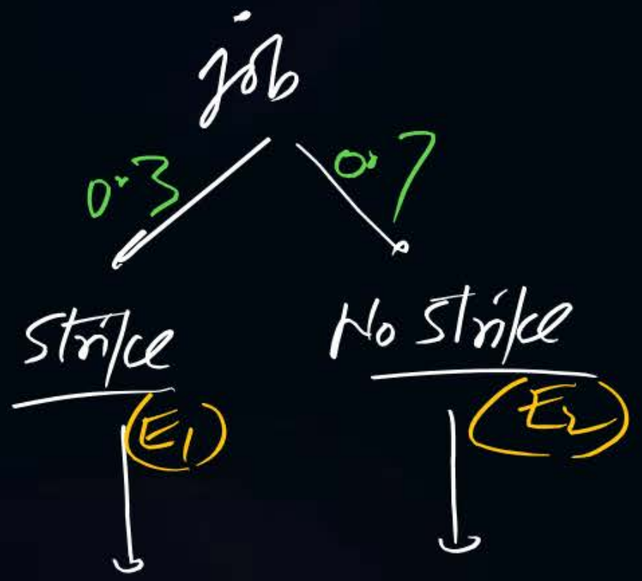
③ 1000 persons → 10% → 100 are Actually +ve → 95% → 95 are declared +ve by kit (True Result)
→ 5% → 5 are " -ve by kit (false ")
→ 90% → 900 " actually -ve → 89% → 801 are declared -ve by kit (True Result)
→ 11% → 99 are " +ve by kit (false ")

Q: A person has undertaken a construction job. The prob that there is a strike is 0.3.
The prob that construction job will be completed on time (if there is a strike) is 0.6
and (if there is No strike) is 0.8
then find the prob that construction job will be completed on time ?

Sol: $A = \{$ Construction job will be completed on time $\}$

$E_1 = \{$ Strike $\} = 0.3$, $A/E_1 = 0.6$

$E_2 = \{$ No Strike $\} = 0.7$ $A/E_2 = 0.8$



$$P(A) = (0.3 \times 0.6) + (0.7 \times 0.8)$$

$$= 0.18 + 0.56$$

$$= 0.74 \simeq 74\% \text{ chance}$$

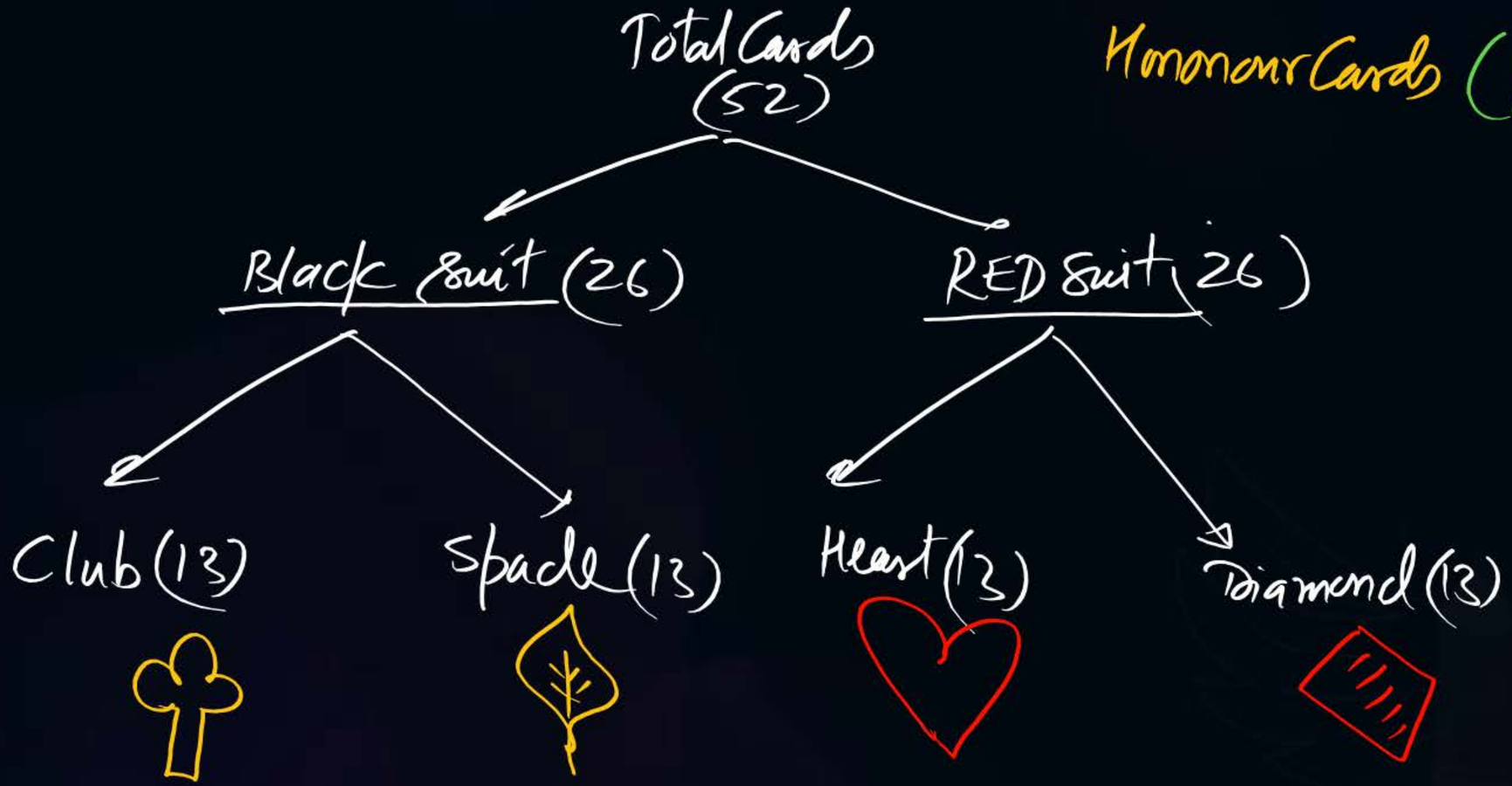
job will be completed on time A :

$$A/E_1 \simeq 0.6 \qquad A/E_2 \simeq 0.8$$

Concept of Playing Cards →

Face Cards (12) eg K, Q, J

Honour Cards (16) eg K, Q, J, A



Q. From a pack of 52 Cards, while shuffling, four cards are accidently dropped. then find the prob that missing cards will be one from each suits ?

Sol: Total ways of missing 4 cards $= {}^{52}C_4$

App II

fav ways of missing 4 cards (one from each suit)

$= {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1$

So Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{({}^{13}C_1)^4}{{}^{52}C_4} = \left(\frac{13\times13\times13\times13}{52\times51\times50\times49}\right) \times 4\times3\times2\times1$

Q. From a pack of 52 cards, while shuffling, one card is dropped (Q-1) & then two cards are drawn at Random (Q-2) then Find the Prob that both the selected cards will be Spade?

Sol: Case I: (Spade is lost) or Case II (Spade Not lost)

$$P(\text{Sp is lost}) = \frac{{}^{13}C_1}{{}^{52}C_1} = \frac{13}{52}$$

Now Remaining Cards: (12 Sp & 39 others)

$$P(\text{Both Cards will Spade}) = \frac{f}{T} = \frac{{}^{12}C_2}{{}^{51}C_2}$$

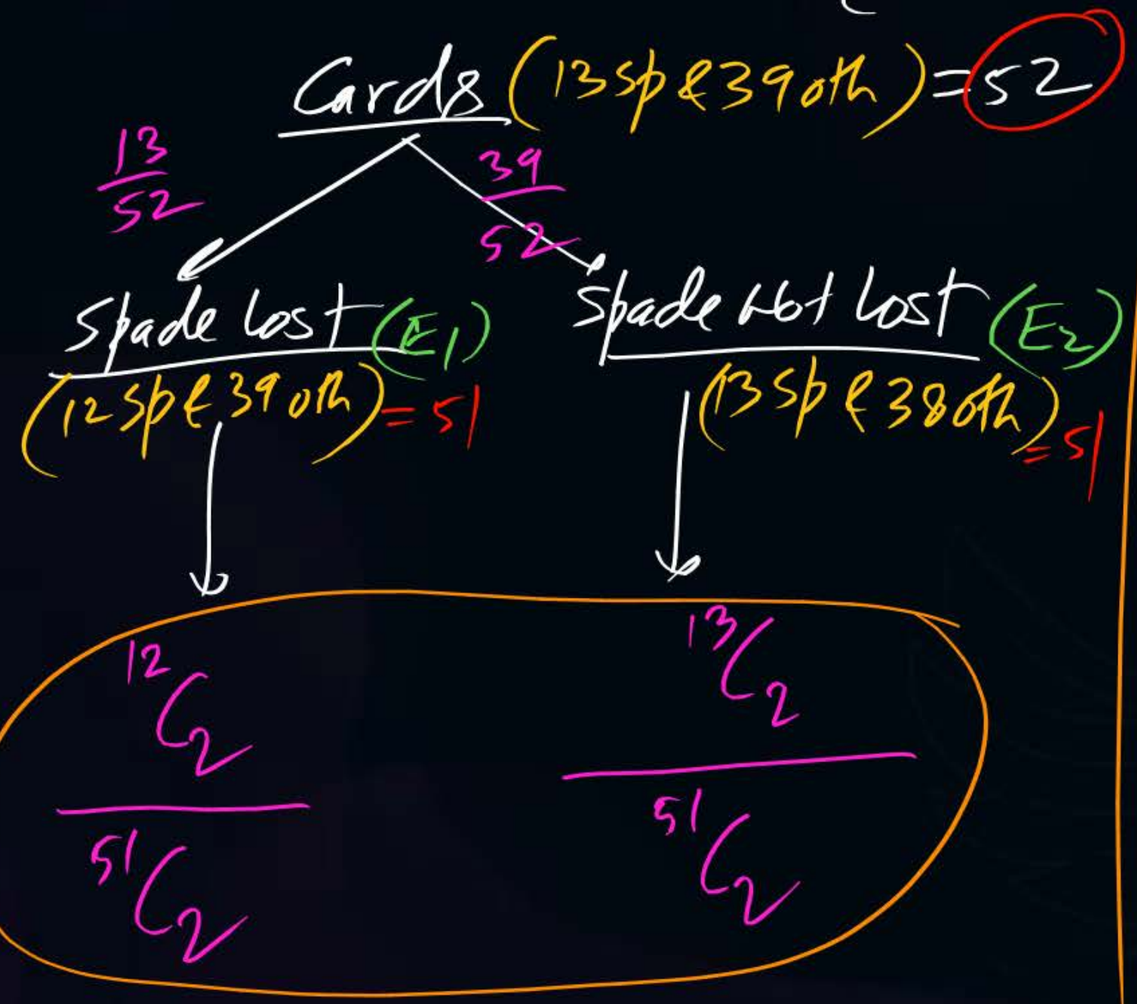
$$P(\text{Sp is not lost}) = \frac{{}^{39}C_1}{{}^{52}C_1} = \frac{39}{52}$$

Now Remaining Cards: (13 Sp & 38 oth)

$$P(\text{Both will be Spade}) = \frac{f}{T} = \frac{{}^{13}C_2}{{}^{51}C_2}$$

$$\text{Req Prob} = \left(\frac{13}{52} \times \frac{{}^{12}C_2}{{}^{51}C_2}\right) + \left(\frac{39}{52} \times \frac{{}^{13}C_2}{{}^{51}C_2}\right) = \frac{1}{4} \times \frac{12 \times 11}{51 \times 50} + \frac{3}{4} \times \frac{13 \times 12}{51 \times 50} = \text{Ans.}$$

M-II Using Law of Total Prob (☺) → A = {Both the selected cards are spade}



$$P(A) = \left(\frac{13}{52} \times \frac{{}^{12}C_2}{{}^{51}C_2}\right) + \left(\frac{39}{52} \times \frac{{}^{13}C_2}{{}^{51}C_2}\right)$$

$$= \frac{1}{17}$$

Note: From a pack of 52 cards while shuffling one card is lost then

(1) find the prob that lost card will spade = ? $= \frac{{}^{13}C_1}{{}^{52}C_1} = \frac{13}{52} = \frac{1}{4}$

(2) " " " " will not spade = ? $= \frac{{}^{39}C_1}{{}^{52}C_1} = \frac{39}{52} = \frac{3}{4}$

Concept of with & W/o Replacement :-

eg: from a pack of 52 cards, 3 cards are drawn then find the Number of Cases if cards are drawn

① Simultaneously = ? $= {}^{52}C_3 = \frac{52 \times 51 \times 50}{3 \times 2 \times 1}$

② one by one with Replacement = ? $= {}^{52}C_1 \times {}^{52}C_1 \times {}^{52}C_1 = 52 \times 52 \times 52$

③ one by one W/o Replacement = ? $= {}^{52}C_1 \times {}^{51}C_1 \times {}^{50}C_1 = 52 \times 51 \times 50$

Eg: From a pack of Regular playing Cards, Two Cards are drawn then Find the Prob that both will be king if 1st Card is Not Replaced.

Sol: Here we are drawing Cards one by one (w/o) Replacement.

(M-I)

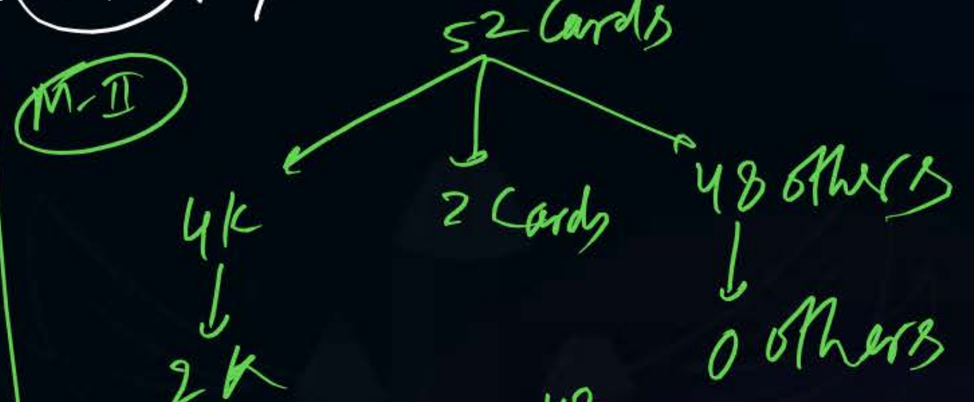
$$\text{Req Prob} = P[K \cap K]$$

$$= P(K \text{ in } 1^{st} \text{ draw}) \times P(K \text{ in } 2^{nd} \text{ draw})$$

$$= \frac{{}^4C_1}{{}^{52}C_1} \times \frac{{}^3C_1}{{}^{51}C_1}$$

$$= \frac{4}{52} \times \frac{3}{51}$$

(M-II)



$$\text{Req Prob} = \frac{f}{T} = \frac{{}^4C_2 \times {}^{48}C_0}{{}^{52}C_2} = \frac{4 \times 3}{52 \times 51}$$

Q2 A Box Contains 4R & 3B Marbles & we want to draw 3 Marbles one by one w/o Replacement then find the Prob of drawing 1R & 2B Marbles ?

M-I (By Making Cases) → (Conceptual App)

Req Prob = P[RBB or BRB or BBR]

$$= \left(\frac{4}{7}\cdot\frac{3}{6}\cdot\frac{2}{5}\right)+\left(\frac{3}{7}\frac{4}{6}\frac{2}{5}\right)+\left(\frac{3}{7}\frac{2}{6}\frac{4}{5}\right)$$

$$= \left(\frac{4}{7}\times\frac{3}{6}\times\frac{2}{5}\right)\times 3$$

Take Care → M-II is applicable only when Q. is Based on W/O Replacement Concept.

M-II (using Hypergeometric Distribution) (Tricky App)



$$\text{Req Prob} = \frac{f}{T} = \frac{{}^4C_1 \times {}^3C_2}{{}^7C_3} = \frac{4\times\frac{3\times2}{2}}{\frac{7\times6\times5}{3\times2}}$$

$$= \left(\frac{4\times3\times2}{7\times6\times5}\right)\times 3$$

(ii) Also find the ans if Marbles are drawn one by one with Replacement.

(M-I) (By Making Cases) → (4R & 3B) → (1R & 2B)

Req Prob = P[RBB or BRB or BBR]

$$= \left(\frac{4}{7}\cdot\frac{3}{7}\cdot\frac{3}{7}\right) + \left(\frac{3}{7}\cdot\frac{4}{7}\cdot\frac{3}{7}\right) + \left(\frac{3}{7}\cdot\frac{3}{7}\cdot\frac{4}{7}\right)$$

$$= \left(\frac{4}{7}\times\frac{3}{7}\times\frac{3}{7}\right)\times 3$$

(M-II) (By BINOMIAL DIST) →

will be discussed Later

Req Prob = P(X = 1R)

$$= {}^{n}C_r\, p^r q^{n-r}$$

$$= {}^{3}C_1 \left(\frac{4}{7}\right)\left(\frac{3}{7}\right)^{3-1}$$

$$= 3\times\left(\frac{4}{7}\times\frac{3}{7}\times\frac{3}{7}\right)$$

Q There are 10 Calci on a Table in which 6 are Defective & 4 are Non Def. & we want to draw three Calci one by one w/o Replacement then Find the Prob that there will be exactly one Defective?

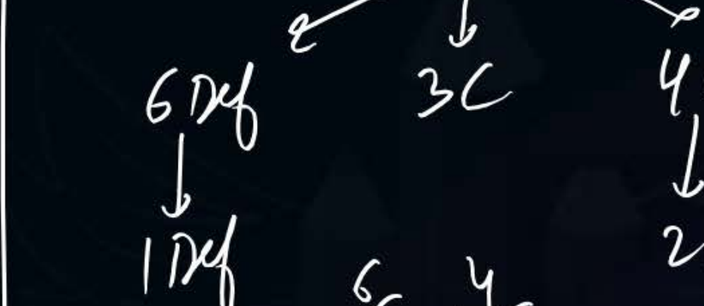
(M-I) Req Prob = P[DNN or NDN or NND]

$$= \left(\frac{6}{10}\frac{4}{9}\frac{3}{8}\right) + \left(\frac{4}{10}\frac{6}{9}\frac{3}{8}\right) + \left(\frac{4}{10}\frac{3}{9}\frac{6}{8}\right)$$

$$= 3 \times \left(\frac{6}{10} \cdot \frac{4}{9} \cdot \frac{3}{8}\right)$$

(M-II) (using Hypergeometric Dist)

10 Calci



$$\text{Req Prob} = \frac{f}{T} = \frac{{}^6C_1 \times {}^4C_2}{{}^{10}C_3} = \frac{6 \times \frac{4\times3}{2}}{\frac{10\times9\times8}{3\times2}}$$

$$= \left(\frac{6}{10} \cdot \frac{4}{9} \cdot \frac{3}{8}\right) \times 3$$

(ii) also find the answer if we are drawing Calci one by one with Replacement?

10 Calci ← 6 Def
4 N.Def

(M-I) Req Prob = P[DNN or NDN or NND]

$$= \left(\frac{6}{10}\frac{4}{10}\frac{4}{10}\right) + \left(\frac{4}{10}\frac{6}{10}\frac{4}{10}\right) + \left(\frac{4}{10}\frac{4}{10}\frac{6}{10}\right)$$

$$= \left(\frac{6}{10} \times \frac{4}{10} \times \frac{4}{10}\right) \times 3$$

(M-II) Using B-Dist.:-

Wait 2 or 3 days.

THANK - YOU